ECE253/CSE208 Introduction to Information Theory

Lecture 7: Entropy Rate

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- Chap 4 of *Elements of Information Theory (2nd Edition)* by Cover–Thomas

Markov Chain (MC)

Consider a discrete-time stochastic process $X_1, X_2, \ldots, X_{n+1}$.

Markov property:

$$\Pr(X_{n+1} = x_{n+1} \mid X_n = x_n, \ldots, X_1 = x_1) = \Pr(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

Equivalently, the joint distribution factorizes as

$$p(x_1, \ldots, x_{n+1}) = p(x_1) \prod_{i=1}^{n} p(x_{i+1} \mid x_i).$$

The current state summarizes all information about the past.

k-th order Markov chain satisfies

$$p(x_{n+1} \mid x_n, \ldots, x_1) = p(x_{n+1} \mid \underbrace{x_n, \ldots, x_{n-k+1}}_{k \text{ terms}}).$$

Time-Homogeneous Markov Chain

Definition (Time-homogeneous MC)

A Markov chain $\{X_n\}_{n\geq 0}$ is **time-homogeneous** if

$$\Pr(X_{n+1} = j \mid X_n = i) = \Pr(X_1 = j \mid X_0 = i) \quad \text{for all } n, i, j.$$

Equivalently,

$$\Pr(X_{n+1} = j \mid X_n = i) = P_{ij},$$

where the state transition matrix $\mathbf{P} = (P_{ij})$ does *not* depend on time n.

n-Step Transition Matrix

- *1-step transition matrix:* $P_{ij} = \Pr(X_{k+1} = j \mid X_k = i)$.
- *n-step transition matrix:* $P_{ij}^{(n)} = \Pr(X_n = j \mid X_0 = i)$.

Fundamental result (for time-homogeneous MC):

$$\boxed{\mathbf{P}^{(n)} = \mathbf{P}^n}$$

Proof (Chapman–Kolmogorov equation reduces to matrix multiplication):

$$P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)}.$$

Contrast: If the chain is *time-inhomogeneous* (transition probabilities depend on time), then generally

$$\mathbf{P}^{(n)} \neq \mathbf{P}^n.$$

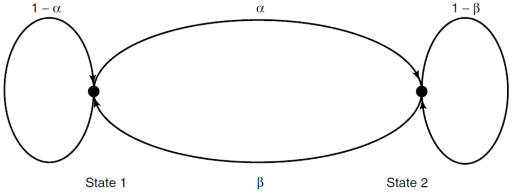
Example: Two-state MC

Example

Consider a two-state MC whose state transition matrix is

$$\mathbf{P} = \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix}$$

The state transitions can be captured by the following diagram:



Evolution of the State Distribution

Consider a finite-state MC with state space $\mathcal{S} := \{1, 2, \ldots, m\}$.
Let the row vector

$$\boldsymbol{\pi}^{(n)} = \left[\pi_1^{(n)}, \pi_2^{(n)}, \ldots, \pi_m^{(n)}\right]$$

denote the **state distribution** at time n, where

$$\pi_i^{(n)} = \Pr(X_n = i), \quad i \in \mathcal{S}.$$

Question. Given $\Pr(X_n = i)$, how do we compute $\Pr(X_{n+1} = j)$?

Answer. By the law of total probability,

$$\Pr(X_{n+1} = j) = \sum_{i \in \mathcal{S}} \Pr(X_n = i) \Pr(X_{n+1} = j \mid X_n = i).$$

Since $\Pr(X_{n+1} = j \mid X_n = i) = P_{ij}$, we obtain

$$\boxed{\boldsymbol{\pi}^{(n+1)} = \boldsymbol{\pi}^{(n)} \mathbf{P}}$$

Stationary Distribution

Definition (Stationary Distribution)

A probability vector $\boldsymbol{\pi}$ is a *stationary distribution* if

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \boldsymbol{\pi} \geq \mathbf{0}, \qquad \boldsymbol{\pi}\mathbf{1} = 1,$$

where $\mathbf{1}$ is the all-ones column vector.

Thus, a stationary distribution is:

- A fixed point of the linear map induced by $\mathbf{P}$,
- A left eigenvector of $\mathbf{P}$ corresponding to eigenvalue 1,
- An element of the probability simplex.

Example of Stationary Distribution

Example

Find the stationary distribution of the aforementioned two-state MC.

$$\begin{cases} \boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi} \\ \boldsymbol{\pi}\mathbf{1} = 1 \end{cases} \quad \Rightarrow \quad \boldsymbol{\pi} = \left[\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta}\right].$$

Limiting Distribution

Definition (Limiting Distribution)

We say a **limiting distribution** exists if

$$\lim_{n\to\infty} P_{ij}^{(n)} = \pi_j \quad \text{for all } i, j,$$

and *the limit is independent of the initial state* i.

In that case, the vector $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_m)$ is called the **limiting distribution**. All rows of $\mathbf{P}^n$ converge to the same vector $\boldsymbol{\pi}$.

Equivalent formulation:

$$\lim_{n\to\infty} \boldsymbol{\mu}\mathbf{P}^n = \boldsymbol{\pi} \quad \text{for any initial distribution } \boldsymbol{\mu}.$$

If a limiting distribution exists, then it must be stationary.

Example of Stationary vs. Limiting Behavior

1. Limiting behavior

$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{P}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{P}^3 = \mathbf{P}.$$

The sequence $\{\mathbf{P}^n\}$ oscillates forever.

$$\boxed{\lim_{n \to \infty} \mathbf{P}^n \text{ does not exist.}}$$

2. Stationary distribution Solve $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi} \implies \boldsymbol{\pi} = \left(\frac{1}{2}, \frac{1}{2}\right)$.

In this example:

- A stationary distribution exists and is unique.
- The limiting distribution does *not* exist.
- Periodicity prevents convergence.

Accessible

Definition (Accessible and Communicate)

- State j is said to be accessible from state i if $P_{ij}^{(n)} > 0$ for some $n \geq 0$, which is denoted as $i \to j$. That is, state j is reachable from state i in n steps.
- Two states i and j are said to communicate if they are accessible from each other, which is denoted as $i \leftrightarrow j$. (i.e., there are directed paths between i and j).

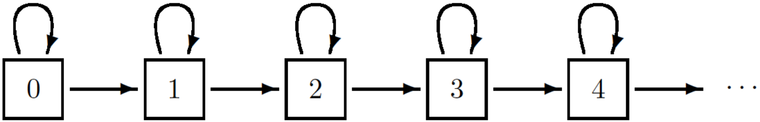


Figure: Each state is accessible from all its previous states, but not vice versa.

Communication Classes

Communication is an *equivalence relation*:

- **Reflexive:** $i \leftrightarrow i$
- **Symmetric:** If $i \leftrightarrow j$, then $j \leftrightarrow i$
- **Transitive:** If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$

Hence, the state space $\mathcal{S}$ is partitioned into disjoint **communication classes**.

Irreducible MC

Definition (Irreducibility)

A Markov chain is **irreducible** if $\forall i, j \in \mathcal{S}, \quad i \to j$.

- The chain is irreducible iff all states belong to a single communication class.
- The chain is reducible if there are two or more communication classes.
- Irreducibility means that for some power n, the matrix $\mathbf{P}^n$ has strictly positive entries.
- Irreducibility is equivalent to the transition graph being strongly connected. That is, every vertex is reachable from every other vertex following the direction of edges.

$$\boxed{\text{Irreducible} \iff \text{Single communication class} \iff \text{Graph is strongly connected}}$$

Example of Irreducible MC

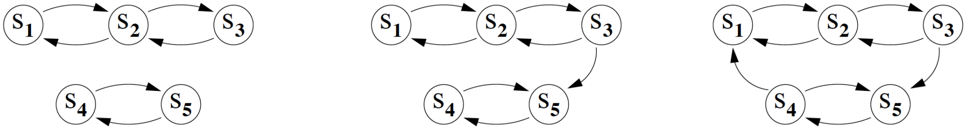


Figure: The first two Markov chains are reducible while the last one is irreducible.

Intuitively, irreducible means:

- No isolated components.
- No closed subgroups.
- The chain behaves as one unified system.

Period and Aperiodicity (1/2)

Definition (Period of a State)

The period of a state i is defined as

$$k(i) = \gcd\left\{n \geq 1 : P_{ii}^{(n)} > 0\right\}^{a}.$$

[a] **gcd** is the greatest common divisor

Interpretation:

- Consider all time steps n at which it is possible to return to state i.
- The period is the greatest common divisor of those return times.

Period and Aperiodicity (2/2)

Aperiodic State if

$$k(i) = 1.$$

Aperiodic Markov Chain:
A Markov chain is aperiodic if every state is aperiodic.

Example:

- If returns occur at times $2, 4, 6, \ldots$ then $k(i) = 2$.
- If returns occur at times $3, 5, 7, \ldots$ then $k(i) = 1$.

Examples of Aperiodic MC

Consider a finite irreducible Markov chain:

- If there is a self transition in the diagram, then the chain is aperiodic.
- Suppose $P_{ii}^{(\ell)} > 0$ and $P_{ii}^{(m)} > 0$. If ℓ and m are co-prime ($\gcd(\ell, m) = 1$), then state i is aperiodic.
- All states in the same communication class have the same period.
- An irreducible MC only needs one aperiodic state to imply the chain is aperiodic.

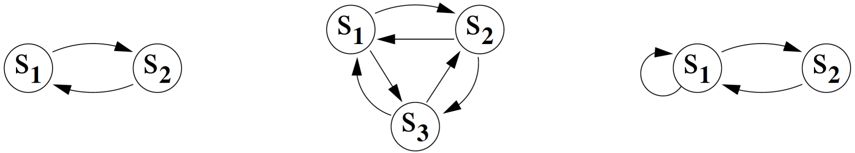


Figure: The first diagram has period 2 while the last two diagrams are aperiodic.

Exercise: Find Communication Classes[1]

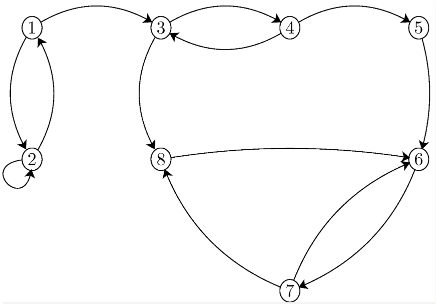


Figure: **Find all communication classes and their periods.**

Answer:

- Class 1={1, 2}, aperiodic
- Class 2={3, 4}, period = 2
- Class 3={5}, period = undefined (the state never returns, it is transient)
- Class 4={6, 7, 8}, aperiodic

[1] https://www.probabilitycourse.com/chapter11/11_2_4_classification_of_states.php

Unique Stationary Distribution and Limiting Distribution (1/2)

Theorem

For an irreducible and aperiodic finite-state MC, there exists a finite integer N such that $P_{ij}^{(n)} > 0$, for all $i, j \in \mathcal{S}$ and all $n \geq N$.

Theorem

A finite irreducible MC has a unique stationary distribution.

Lemma

For an irreducible and aperiodic finite-state MC, any initial distribution converges to the unique stationary distribution.

Unique Stationary Distribution and Limiting Distribution (2/2)

Abbreviations: Stationary (Stat.), Irreducible (Irr.), Aperiodic (Aper.)

Assumptions	**Stat. Exists**	**Stat. Unique**	**Limit Exists**	**Limit = Stat.**
Finite	Yes	Not necessarily	Not necessarily	Not necessarily
Finite + Irr.	Yes	Yes	Not necessarily	If aperiodic
Finite + Irr. + Aper.	Yes	Yes	Yes	Yes

Return Probability and Expected Return Time

First Return Time

For a Markov chain $\{X_n\}_{n\geq 0}$, define the first return time to state i:

$$T_i = \inf\{n \geq 1 : X_n = i\}.$$

Return Probability

$$\Pr_i(T_i < \infty) = \Pr(\exists\, n \geq 1 \text{ such that } X_n = i \mid X_0 = i)\,.$$

Expected Return Time

$$\mathbb{E}_i[T_i] = \mathbb{E}[\inf\{n \geq 1 : X_n = i\} \mid X_0 = i]\,.$$

Recurrent vs Transient

Type	Return Prob	Expected Return Time	Stationary Distribution
Transient	< 1	—	No
Null recurrent	$= 1$	$= \infty$	No
Positive recurrent	$= 1$	$< \infty$	Yes (unique)

Fundamental Theorem

For an irreducible countable-state Markov chain:

$$\text{Stationary distribution exists} \iff \text{Chain is positive recurrent.}$$

Remark: Finite-state chains are automatically positive recurrent.

Markov Chain for English Texts[2] (1/2)

- **Zero-order approx.** *(symbols independent and equiprobable).*
  XFOML RXKHRJFFJUJ ZLPWCFWKCYJ FFJEYVKCQSGHYD QPAAMKBZAACIBZL HJQD.
- **First-order approx.** *(symbols independent but with frequencies of English text).*
  OCRO HLI RGWR NMIELWIS EU LL NBNESEBYA TH EEI ALHENHTTPA OOBTTVANAH BRL.
- **Second-order approx.** *(bigram structure: next letter depends only on the previous letter).*
  ON IE ANTSOUTINYS ARE T INCTORE ST BE S DEAMY ACHIN D ILONASIVE TUCOOWE AT TEASONARE FUSO TIZIN ANDY TOBE SEACE CTISBE.
- **Third-order approx.** *(trigram structure: next letter depends on the previous two letters).*
  IN NO IST LAT WHEY CRATICT FROURE BIRS GROID PONDENOME OF DEMONSTURES OF THE REPTAGIN IS REGOACTIONA OF CRE.

[2] Section 3 of the paper *"A Mathematical Theory of Communication (1948)"*

Markov Chain for English Texts (2/2)

- **First-order word approx.** *(words are chosen independently with appropriate frequencies).* REPRESENTING AND SPEEDILY IS AN GOOD APT OR COME CAN DIFFERENT NATURAL HERE HE THE A IN CAME THE TO OF TO EXPERT GRAY COME TO FURNISHES THE LINE MESSAGE HAD BE THESE.
- **Second-order word approx.** *(correct word transition probabilities but without further structure).*
  THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHARACTER OF THIS POINT IS THEREFORE ANOTHER METHOD FOR THE LETTERS THAT THE TIME OF WHO EVER TOLD THE PROBLEM FOR AN UNEXPECTED.

Shannon asked:
"Can we define a quantity which will measure, in some sense, how much information is "produced" by such a process, or better, at what rate information is produced?"

Stationary Stochastic Process

Discrete-time Information Sources:

- Communications take place continually rather than a finite period of time; e.g., Internet cellular networks, ratio stations, TV program, etc.
- The info source can be modeled as a discrete-time stochastic process $\{X_k\}_{k=1}^{\infty}$

Definition (Stationary stochastic process)

A stochastic process $\{X_k\}$ is *strongly stationary* if the joint distribution of any subset of the sequence is invariant w.r.t. time shifts. That is,

$$\Pr\{X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n\} = \Pr\{X_{1+l} = x_1, X_{2+l} = x_2, \ldots, X_{n+l} = x_n\}$$

for every n, every shift l, and for all $x_1, \ldots, x_n \in \mathcal{X}$.

MC May Not Be Stationary

Q: Is any Markov chain a stationary stochastic process? **A:** No.

Example

Consider an MC with $\mathbf{P} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$ and the initial probability state $\boldsymbol{\pi}^{(0)} = (1, 0)$.

Then, $\boldsymbol{\pi}^{(1)} = \boldsymbol{\pi}^{(0)} \times \mathbf{P} = (1/2, 1/2) \neq \boldsymbol{\pi}^{(0)} \implies$ not stationary.

Entropy Rate for Stochastic Processes

Q: How does the entropy of a sequence grow with n?

A: We define the entropy rate as the rate of growth:

Definition (Per symbol entropy of n random variables)

$$H(\mathcal{X}) := \lim_{n\to\infty} \frac{1}{n} H(X_1^n)$$

when the limit exists.

Special cases:

- $\{X_i\}$ are i.i.d.: $H(\mathcal{X}) = H(X_1)$.
- $\{X_i\}$ are independent but not identical: $H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} H(X_i)$. But, the limit may not even exist; e.g, $H(X_i) = i$.

Alternative Definition of Entropy Rate

Definition (Conditional entropy of the last random variable given the past.)

$$H'(\mathcal{X}) := \lim_{n\to\infty} H(X_n|X_1^{n-1})$$

Entropy Rate for Stationary Stochastic Processes

Theorem (Entropy rate)

For a stationary stochastic process, $H(\mathcal{X}) = H'(\mathcal{X})$.

Proof: Due to the stationarity of the process, we have

$$0 \leq H(X_n|X_1^{n-1}) \leq H(X_n|X_2^{n-1}) = H(X_{n-1}|X_1^{n-2}),$$

Therefore, $H(X_n|X_1^{n-1})$ is a monotonically non-increasing sequence and lower bounded by 0. Hence, the sequence must converge $\lim_{n\to\infty} H(X_n|X_1^{n-1}) = H'(\mathcal{X})$. Recall that $\frac{1}{n}H(X_1,\ldots,X_n) = \frac{1}{n}\sum_{i=1}^n H(X_i|X_1^{i-1})$. By the lemma of Cesáro mean, the RHS converges to $H'(\mathcal{X})$, and so does the LHS.

Lemma (Cesáro mean)

If a sequence $\{a_n\} \to c$*, the running average* $\{b_n := \frac{1}{n}\sum_{i=1}^n a_i\} \to c$.

Ergodicity and Entropy Rate

Definition (Ergodic Markov Chain)

A finite-state Markov chain is *ergodic* if it is: *Irreducible* and *Aperiodic*

Ergodic Theorem for Finite MC: Time average = Ensemble average (almost surely)

If $\{X_n\}$ is irreducible and aperiodic with stationary distribution π, then for any function f,

$$\frac{1}{n}\sum_{k=1}^{n} f(X_k) \xrightarrow{a.s.} \mathbb{E}_\pi[f(X)].$$

Implication for Entropy Rate

For stationary ergodic processes,

$$-\frac{1}{n}\log P(X_1^n) \xrightarrow{a.s.} H(X).$$

This justifies interpreting entropy rate as the *asymptotic uncertainty per symbol*.

Existence of Entropy Rate via Subadditivity

Key Question: Does $\lim_{n\to\infty} \frac{1}{n} H(X_1^n)$ exist?

Subadditivity Property:
For any stochastic process, $H(X_1^{m+n}) \le H(X_1^m) + H(X_{m+1}^{m+n})$.
Hence $\{H(X_1^n)\}$ is a subadditive sequence.

Fekete's Lemma:
If $\{a_n\}$ is subadditive, then $\frac{a_n}{n} \to \inf_n \frac{a_n}{n}$.

Conclusion:
The entropy rate always exists as $H(\mathcal{X}) = \inf_n \frac{1}{n} H(X_1^n)$.
Stationarity is only needed to connect this with the conditional entropy form.

Entropy Rate for Stationary MC

Lemma

For a stationary Markov chain, $H'(\mathcal{X}) = H(X_2|X_1)$.

Proof: $H'(\mathcal{X}) = \lim_{n\to\infty} H(X_n|X_1^{n-1}) = \lim_{n\to\infty} H(X_n|X_{n-1}) = H(X_2|X_1)$.

Theorem

Let $\{X_i\}$ *be a stationary Markov chain with stationary distribution* $\boldsymbol{\mu}$ *and transition probability matrix* $\mathbf{P}$. *If* $X_1 \sim \boldsymbol{\mu}$, $H(\mathcal{X}) = -\sum_{ij} \mu_i P_{ij} \log P_{ij}$

Proof: $H(\mathcal{X}) = H(X_2|X_1) = \sum_i \Pr(X_1 = i) H(X_2|X_1 = i) = \sum_i \mu_i \sum_j P_{ij} \log P_{ij}^{-1}$.

Note: For a finite-state, irreducible, and aperiodic Markov chain, the entropy rate $H(\mathcal{X})$ does not depend on the choice of the initial distribution, and it is not necessary to start the chain in stationarity.

Entropy Rate of Random Walk over Graph (1/3)

Example (Random walk over a weighted graph)

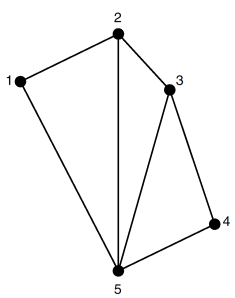


Figure: Consider the simple random walk on a finite, connected, undirected, weighted graph $G(\mathcal{N}, \mathcal{E}, \mathcal{W})$.

- $w_{ij} = w_{ji}$ denotes the edge weight between nodes i and j (0 if no edges).
- Given $X_n = i$, the probability of moving from node i to j is $P_{ij} = \frac{w_{ij}}{\sum_k w_{ik}} = \frac{w_{ij}}{w_i}$, where $w_i := \sum_k w_{ik}$ is the total weight of all edges connecting with node i.

Entropy Rate of Random Walk over Graph (2/3)

Example (Random walk over a weighted graph)

- Intuitively, the stationary distribution of any node $i \in \mathcal{N}$ should be proportional to its degree w_i, which can be derived as $\pi_i = \frac{w_i}{2w}$, where $w \triangleq \sum_{i,j:j>i} w_{ij}$ is the total weight of all edges.

- Sanity check of the stationary distribution:

$$\sum_i \pi_i P_{ij} = \sum_i \frac{w_i}{2w} \frac{w_{ij}}{w_i} = \frac{w_j}{2w} = \pi_j, \ \forall j \in \mathcal{N}.$$

- Locality property of this stationary distribution: it depends only on the total weight and the weight of edges connected to the node.

Entropy Rate of Random Walk over Graph (3/3)

Hence, the entropy rate is

$$
\begin{aligned}
H(\mathcal{X}) = H(X_2|X_1) &= -\sum_{ij} \mu_i P_{ij} \log P_{ij} &\quad (1)\\
&= -\sum_{ij} \frac{w_{ij}}{2w} \log \frac{w_{ij}}{w_i} &\quad (2)\\
&= -\sum_{ij} \frac{w_{ij}}{2w} \log \frac{w_{ij}}{2w} + \sum_i \frac{w_i}{2w} \log \frac{w_i}{2w} &\quad (3)\\
&= \underbrace{H\left(\ldots, \frac{w_{ij}}{2w}, \ldots\right)}_{|\mathcal{N}|^2 \text{ terms}} - \underbrace{H\left(\ldots, \frac{w_i}{2w}, \ldots\right)}_{|\mathcal{N}| \text{ terms}} &\quad (4)
\end{aligned}
$$

If all the edges have equal weight, the stationary distribution becomes $\pi_i = \frac{D_i}{2D}$, where D_i is the degree of node i and D is the total degree of the graph. In this case, the entropy rate is

$$H(\mathcal{X}) = \log(2D) - H\left(\frac{D_1}{2D}, \ldots, \frac{D_{|\mathcal{N}|}}{2D}\right).$$

Function of Markov Chain (1/3)

Theorem

Consider a stationary Markov chain $\{X_i\}$ and $Y_i = \phi(X_i)$ for all i. We have:

$$H(Y_n|Y_1^{n-1}, X_1) \leq H(\mathcal{Y}) \leq H(Y_n|Y_1^{n-1})$$

$$\lim_{n\to\infty} H(Y_n|Y_1^{n-1}, X_1) = H(\mathcal{Y}) = \lim_{n\to\infty} H(Y_n|Y_1^{n-1}).$$

First, note that $\{X_i\}$ is a stationary MC $\implies$ $\{Y_i\}$ is stationary, but not necessarily a MC (unless ϕ is injective).

$$\begin{aligned}
\Pr\left(Y_{n+1} = y_{n+1}|\{Y_k = y_k\}_{k\leq n}\right) &= \Pr\left(X_{n+1} = \phi^{-1}(y_{n+1})|\{X_k = \phi^{-1}(y_k)\}_{k\leq n}\right) && (5)\\
&= \Pr\left(X_{n+1} = \phi^{-1}(y_{n+1})|X_n = \phi^{-1}(y_n)\right) && (6)\\
&= \Pr\left(Y_{n+1} = y_{n+1}|Y_n = y_n\right) && (7)
\end{aligned}$$

Function of Markov Chain (2/3)

Proof: We have proved the upper bound. For the lower bounded,

$$
\begin{aligned}
H(Y_n|Y_1^{n-1}, X_1) &= H(Y_n|Y_1^{n-1}, X_1, X_0^{-k}) && (8)\\
&= H(Y_n|Y_1^{n-1}, X_1, X_0^{-k}, Y_{-k}^0) && (9)\\
&\leq H(Y_n|Y_1^{n-1}, Y_{-k}^0) && (10)\\
&= H(Y_{n+k+1}|Y_1^{n+k}) && (11)
\end{aligned}
$$

The inequality is true for all k, it is also true in the limit.

$$H(Y_n|Y_1^{n-1}, X_1) \leq \lim_{k\to\infty} H(Y_{n+k+1}|Y_1^{n+k}) = H(\mathcal{Y}).$$

Function of Markov Chain (3/3)

Next, we show that

$$\lim_{n\to\infty} H(Y_n|Y_1^{n-1}, X_1) = \lim_{n\to\infty} H(Y_n|Y_1^{n-1}),$$

which is equivalent to $\lim_{n\to\infty} I(X_1; Y_n|Y_1^{n-1}) = 0$.

Note that $\underbrace{I(X_1; Y_1^n)}_{\text{increases in } n} = H(X_1) - H(X_1|Y_1^n) \leq H(X_1)$. Hence, we have

$$H(X_1) \geq \lim_{n\to\infty} I(X_1; Y_1^n) = \lim_{n\to\infty} \sum_{i=1}^{n} I(X_1; Y_i|Y_1^{i-1}) = \sum_{i=1}^{\infty} I(X_1; Y_i|Y_1^{i-1}) \tag{12}$$

The infinite sum of nonnegative terms is finite $\implies$ the terms must tend to 0.

Hidden Markov Model (HMM)[3]

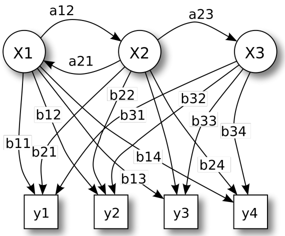


Figure: HMM diagram: $X \to$ states; $y \to$ possible observations; $a \to$ state transition probabilities; $b \to$ output (or emission) probabilities.

Given a Markov process $\{X_n\}$, each Y_i is drawn according to $p(y_i|x_i)$, conditionally independent of all the other X_j, $j \neq i$; i.e.,

$$\boxed{p(x^n, y^n) = p(x^n)p(y^n|x^n) = p(x_1)\prod_{i=1}^{n-1} p(x_{i+1}|x_i)\prod_{i=1}^{n} p(y_i|x_i)}$$

Entropy Rate of Hidden Markov Models

Let $\{X_n\}$ be a stationary Markov chain. Let $\{Y_n\}$ be generated conditionally independently given X_n via $p(y_n \mid x_n)$.

Entropy Rate Bounds.
Let $H(\mathcal{X})$ and $H(\mathcal{Y})$ denote entropy rates. We have

$$H(\mathcal{Y} \mid \mathcal{X}) \leq H(\mathcal{Y}) \leq H(\mathcal{X}) + H(\mathcal{Y} \mid \mathcal{X}).$$

In general, $H(\mathcal{Y})$ has no closed-form expression. Its computation typically involves:

- Forward recursion, random matrix products, Lyapunov exponents

The entropy rate of HMMs remains an active research topic.

Take-away of Entropy Rate

Special Cases

- i.i.d. process:

$$H(\mathcal{X}) = H(X_1).$$

- Stationary process:

$$H(\mathcal{X}) = \lim_{n\to\infty} H(X_n \mid X_1^{n-1}).$$

- Stationary Markov chain:

$$H(\mathcal{X}) = -\sum_{i,j} \pi_i P_{ij} \log P_{ij}.$$

Key Insights

- Entropy rate measures uncertainty growth per symbol.
- Long memory structures reduce entropy rate.
- For ergodic processes:

$$-\frac{1}{n} \log P(X_1^n) \to H(X).$$